

where $\Delta_F(x - y)$ is the scalar field propagator, calculated in the next section. There are no terms of third order in $\mathcal{H}(x)$, or of any odd order in $\mathcal{H}(x)$.

6.2 Calculation of the Propagator

We now turn to a calculation of the propagator (6.1.14), an essential ingredient in the Feynman rules that arises in the pairing of a field $\psi_\ell(x)$ with a field adjoint $\psi_m^\dagger(y)$. Inserting Eqs. (6.1.15) and (6.1.16) in Eq. (6.1.14), and using the commutation or anticommutation relations for annihilation and creation operators, we have immediately

$$\begin{aligned} -i\Delta_{\ell m}(x, y) = & \theta(x - y)(2\pi)^{-3/2} \int d^3p \sum_{\sigma} u_{\ell}(\mathbf{p} \sigma n) u_m^*(\mathbf{p} \sigma n) e^{ip \cdot (x - y)} \\ & \pm \theta(y - x)(2\pi)^{-3/2} \int d^3p \sum_{\sigma} v_m^*(\mathbf{p} \sigma n) v_{\ell}(\mathbf{p} \sigma n) e^{ip \cdot (y - x)} . \end{aligned} \quad (6.2.1)$$

In the course of calculating commutators and anticommutators in Chapter 5, we showed that

$$\sum_{\sigma} u_{\ell}(\mathbf{p} \sigma n) u_m^*(\mathbf{p} \sigma n) = \left(2\sqrt{\mathbf{p}^2 + m_n^2}\right)^{-1} P_{\ell m} \left(\mathbf{p}, \sqrt{\mathbf{p}^2 + m_n^2}\right) , \quad (6.2.2)$$

$$\sum_{\sigma} v_{\ell}(\mathbf{p} \sigma n) v_m^*(\mathbf{p} \sigma n) = \pm \left(2\sqrt{\mathbf{p}^2 + m_n^2}\right)^{-1} P_{\ell m} \left(-\mathbf{p}, -\sqrt{\mathbf{p}^2 + m_n^2}\right) , \quad (6.2.3)$$

where $P_{\ell m}(\mathbf{p}, \omega)$ is a polynomial in $\mathbf{p}$ and ω . (Here as in Eq. (6.2.1), the top and bottom signs refer to bosonic and fermionic fields, respectively.) For instance, if $\psi_{\ell}(x)$ and $\psi_m(y)$ are scalar fields $\phi(x)$ and $\phi(y)$ for a particle of spin zero, then we have simply

$$P(p) = 1 . \quad (6.2.4)$$

If $\psi_{\ell}(x)$ and $\psi_m(y)$ are Dirac fields for a particle of spin $\frac{1}{2}$, then

$$P_{\ell m}(p) = \left[(-i\gamma_{\mu} p^{\mu} + m)\beta\right]_{\ell m} , \quad (6.2.5)$$

where ℓ and m are here four-valued Dirac indices. (The matrix β appears here because we are considering the pairing of $\psi_{\ell}(x)$ with $\psi_m^{\dagger}(y)$. It is absent in the pairing of $\psi(x)$ with $\bar{\psi}(y) \equiv \psi^{\dagger}(y)\beta$.) If $\psi_{\ell}(x)$ and $\psi_m(y)$ are vector fields $V_{\mu}(x)$ and $V_{\nu}(y)$ for a particle of spin one, then

$$P_{\mu\nu}(p) = \eta_{\mu\nu} + m^{-2} p_{\mu} p_{\nu} . \quad (6.2.6)$$